

# Where Does Being Early Pay?

## A catalogue of testable hypotheses for Paper B

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Each hypothesis below says where the penalty for leading language should be *smaller* (or reversed into an advantage) and where it should be *larger*. The outcome throughout is the one the paper already uses — failure at the five-year proof of continued use, plus the capital-markets and acquisition rungs where noted — and every test is the same comparison: the most leading fifth against the most lagging fifth, within Nice class and year, estimated separately inside the groups the hypothesis names and compared across them. “Ready” means the data exist in the repository; “build” means a new variable must be constructed from data we hold; “new data” means a source we do not yet have.

### Where each literature looked, and who answers whom

| Work                                    | Setting                                                                                                                  | Cites, among these works    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Robinson & Fornell (1985)               | Surveyed consumer-goods businesses (market share by order of entry)                                                      | no reference list deposited |
| Teece (1986)                            | Named innovations and their imitators (theory with cases)                                                                | no reference list deposited |
| Lieberman & Montgomery (1988)           | Catalogue of mechanisms across industries                                                                                | Robinson & Fornell; Teece   |
| Golder & Tellis (1993)                  | Histories of fifty product categories, including diapers, video recorders, personal computers, light beer, safety razors | no reference list deposited |
| Klepper & Miller (1995); Klepper (1997) | New manufactured products through their shakeouts                                                                        | no reference list deposited |
| Lieberman & Montgomery (1998)           | Retrospective; entry timing as a question of resources                                                                   | no reference list deposited |

| Work                       | Setting                                                                                       | Cites, among these works                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Kim & Mauborgne (2004)     | Cases of firms that opened new categories (wine, fitness, circus, air travel, hotels, coffee) | cited by neither of the later works with deposited lists                         |
| Suárez & Lanzolla (2007)   | Reconciliation by pace of technology and market change                                        | Robinson & Fornell; Lieberman & Montgomery 1988 and 1998; Golder & Tellis; Teece |
| Semadeni & Anderson (2010) | Consulting firms' service-mark filings (imitation of new offerings)                           | Lieberman & Montgomery 1988 and 1998; Suárez & Lanzolla                          |

The pioneer-advantage literature answers itself in sequence; the blue-ocean literature runs beside it. Links come from the reference lists publishers deposit with Crossref, so a blank means the list is not deposited, not that the citation is absent. The settings suggest where each claim should hold if it holds anywhere: consumer packaged goods for the pioneer claim, the named histories and shakeout-prone manufactured goods for the counter-claim, and the blue-ocean exemplars' own verticals (hypotheses 17–21).

## A. First-mover mechanisms (Lieberman & Montgomery, 1988)

|   | Claim                                                                                              | Test in the record                                                                                                                                                                   | Data  |
|---|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | <i>Technological leadership.</i> Early entrants protected by patents and learning keep their lead. | Lead penalty for owners who also patent (PatentsView match) against those who do not; expect smaller for patenters.                                                                  | ready |
| 2 | <i>Preemption of scarce assets.</i> Early entrants take the best locations and inputs.             | Themes built on physical sites (stores, restaurants, hotels, clinics; classes 35, 43, 44 location vocabulary) against footloose services; expect smaller penalty where sites matter. | build |
| 3 | <i>Buyer switching costs.</i> Customers locked in by contracts stay with the pioneer.              | Descriptions naming subscriptions, memberships, accounts or contracts against one-off goods; expect smaller penalty.                                                                 | build |
| 4 | <i>Network effects.</i> An installed base compounds.                                               | Platform and marketplace vocabulary (platform, marketplace, social network, exchange, community); expect smaller penalty, possibly a reward.                                         | build |

|   | Claim                                                                                        | Test in the record                                                                                                                                                        | Data  |
|---|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5 | <i>Free-riding.</i> Followers copy cheaply what pioneers proved.                             | Themes with low capital intensity and easy imitation (apparel, simple consumer goods, personal services) against capital-heavy manufactured goods; expect larger penalty. | build |
| 6 | <i>Resolution of uncertainty.</i> Pioneers bet before the market settles.                    | Themes whose later share was volatile (surged then receded) against themes that grew steadily; expect larger penalty in volatile themes.                                  | ready |
| 7 | <i>Incumbent inertia.</i> Established firms are slow to follow, so the pioneer's lead lasts. | Lead penalty for debut owners against owners with large existing portfolios filing the same leading language; expect the penalty to fall on new firms.                    | ready |

## B. Pace of change (Suárez & Lanzolla, 2007)

|    | Claim                                                                                                     | Test in the record                                                                                                   | Data  |
|----|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------|
| 8  | <i>Calm waters</i> (slow technology, slow market): a first-mover advantage is durable.                    | Classes and eras with low turnover in the theme mix and flat filing volume; expect the smallest penalty or a reward. | build |
| 9  | <i>Technology leads</i> (fast technology, slow market): early entrants are overtaken by later technology. | Fast theme turnover with flat volume; expect the largest penalty.                                                    | build |
| 10 | <i>Market leads</i> (slow technology, fast market growth): early entrants capture the growing demand.     | Surging filing volume with a stable theme mix (consumer categories, beverages); expect a smaller penalty.            | build |
| 11 | <i>Rough waters</i> (both fast): advantage only for resource-rich entrants.                               | Both fast (technology classes 1995–2000); expect a large penalty, smaller for large or funded owners.                | build |

Pace of technology: year-to-year divergence in a class's theme mix. Pace of market: growth in the class's (or theme's) filing volume. The two cut the class-years into the four quadrants.

## C. Appropriability (Teece, 1986)

|    | Claim                                                                                                                | Test in the record                                                                                                                                                         | Data  |
|----|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 12 | <i>Tight appropriability</i> : where innovations can be protected, the innovator profits.                            | Pharmaceuticals and medical devices (classes 5, 10) and patenting owners; expect smaller penalty.                                                                          | ready |
| 13 | <i>Weak appropriability, generic complementary assets</i> : imitators win.                                           | Services and software with no patenting; expect larger penalty.                                                                                                            | ready |
| 14 | <i>Specialized complementary assets held by incumbents</i> : incumbents who own distribution capture the innovation. | In distribution-heavy consumer goods (food, beverages, cosmetics), the interaction of lead with owner size: leading small entrants fail, leading large incumbents survive. | ready |

#### D. Pioneer disadvantage (Golder & Tellis, 1993) and shakeouts (Klepper, 1997; Klepper & Miller, 1995)

|    | Claim                                                                                                                       | Test in the record                                                                                                                                                          | Data  |
|----|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 15 | <i>Early leaders beat pioneers</i> : the firms remembered as pioneers usually entered a few years after the first entrants. | Within each theme's arrival in a class, the first year's filers against those two to five years in; expect the later cohort to survive better.                              | ready |
| 16 | <i>Survivor bias</i> : pioneer advantage appears only among survivors.                                                      | The lead gradient among all applications, among registrations, and among marks that passed the proof; expect the gradient to fade as the sample is conditioned on survival. | ready |
| 17 | <i>Pioneer advantage in consumer packaged goods</i> (the finding Golder and Tellis dispute).                                | Consumer packaged goods classes (3, 29, 30, 32, 33) against industrial goods; the pioneer literature predicts opposite signs.                                               | ready |
| 18 | <i>Early entrants win shakeouts</i> in manufactured products with scale economies.                                          | Manufactured goods classes (7, 9, 12) against services; and, within surging themes, early joiners against late joiners (already: no order effect).                          | ready |
| 19 | <i>The cited cases</i> : disposable diapers, video recorders, personal computers, light beer, safety razors.                | Lead penalty within the verticals those cases come from (paper and sanitary goods, consumer electronics, beer, razors and blades).                                          | ready |

#### E. Blue ocean (Kim & Mauborgne, 2004)

|    | Claim                                                                                                                                                                               | Test in the record                                                                                                                                                            | Data  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 20 | <i>Reconstructing market boundaries</i> : value comes from combining attributes from different industries.                                                                          | Filings carrying a theme established in another class into their own (new-to-class), and filings whose leading language is a new pairing of existing themes; expect a reward. | ready |
| 21 | <i>The exemplars’ verticals</i> : wine ([yellow tail]), fitness (Curves), live entertainment (Cirque du Soleil), air travel (Southwest, NetJets), hotels, coffee, home improvement. | Lead penalty within each vertical’s class and themes; and where the exemplar’s own filings fall in the lead distribution when they are in the scored years.                   | ready |
| 22 | <i>Creating new demand</i> (non-customers) rather than taking share.                                                                                                                | Themes whose growth came from first-time filers rather than established owners; expect a smaller penalty for the early entrants of such themes.                               | build |

## F. Kinds of new

|    | Claim                                                                                                                                     | Test in the record                                                                                                                                                      | Data  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 23 | <i>New inventions</i> reward pioneers who define the category.                                                                            | Filings carrying a theme new to the whole corpus (too few at 500 themes; use curated invention vocabularies: 3D printing, drones, gene therapy, electric vehicles).     | build |
| 24 | <i>Fads</i> punish late entry, and possibly early entry too.                                                                              | Themes that surged and then fell back below their pre-surge share within five years, against surges that held.                                                          | ready |
| 25 | <i>General-purpose technologies</i> (internet, mobile, cloud, AI) spread across many industries; early adopters in each industry may win. | Vocabularies appearing in ten or more classes; the lead penalty inside each adopting class, by years since the vocabulary reached it (the internet result generalized). | ready |
| 26 | <i>New to the class versus new to the world</i> . Importing an established theme is safer than originating one.                           | New-to-class against new-to-corpus themes, and both against established themes.                                                                                         | ready |
| 27 | <i>Groups of emergent consumer themes</i> (plant-based milk, kombucha, cold brew, hard seltzer, CBD, energy drinks).                      | Pooled into one “emergent consumer category” group, with each category’s early entrants compared with its later entrants and with the class.                            | ready |

## G. Who and where

|    | Claim                                                                                                                                                       | Test in the record                                                                                                   | Data               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------|
| 28 | <i>Clusters speed imitation.</i> Where a theme’s early filers are geographically concentrated (a Silicon Valley or Silicon Alley), followers learn fastest. | Share of a theme’s first five years of filers in its top state; expect a larger penalty in concentrated themes.      | build (state only) |
| 29 | <i>Distance from the cluster.</i> Early filers far from a theme’s home state are shielded from local imitation, or cut off from its resources.              | Within theme, filers in the theme’s hub state against filers elsewhere; the sign is the question.                    | build (state only) |
| 30 | <i>Foreign pioneers.</i> Foreign firms bringing a theme into the US market face local imitators.                                                            | Foreign-domiciled and Madrid filers against US filers.                                                               | ready              |
| 31 | <i>Resources absorb the penalty.</i>                                                                                                                        | Counsel-represented and large-portfolio owners against self-filed debut owners.                                      | ready              |
| 32 | <i>Speculation.</i> Intent-to-use filings claim a market before entering it.                                                                                | Intent-to-use against use-based filings; expect a larger penalty for speculative leading filings.                    | ready              |
| 33 | <i>Regulation as a barrier.</i> Approval requirements protect the first entrant.                                                                            | Regulated classes (pharmaceuticals, medical devices, alcohol, tobacco, financial services) against unregulated ones. | ready              |
| 34 | <i>Timing of entry in the cycle.</i> Early entry in a boom is punished; in a bust it is rewarded.                                                           | The era swing, extended by the reversal decomposition to every class and theme.                                      | ready              |

## Notes on design

Many of these groupings overlap, and several are proxies (a vocabulary for “switching costs” is not a measured switching cost). The battery should be run with one pre-stated prediction per hypothesis, a common estimator, owner-clustered errors, and a correction for testing thirty comparisons at once; the paper would then report where the evidence lines up with each literature, not a search for significance. City-level geography would need owner addresses re-parsed from the raw case files; the state-level version is ready.

## References

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